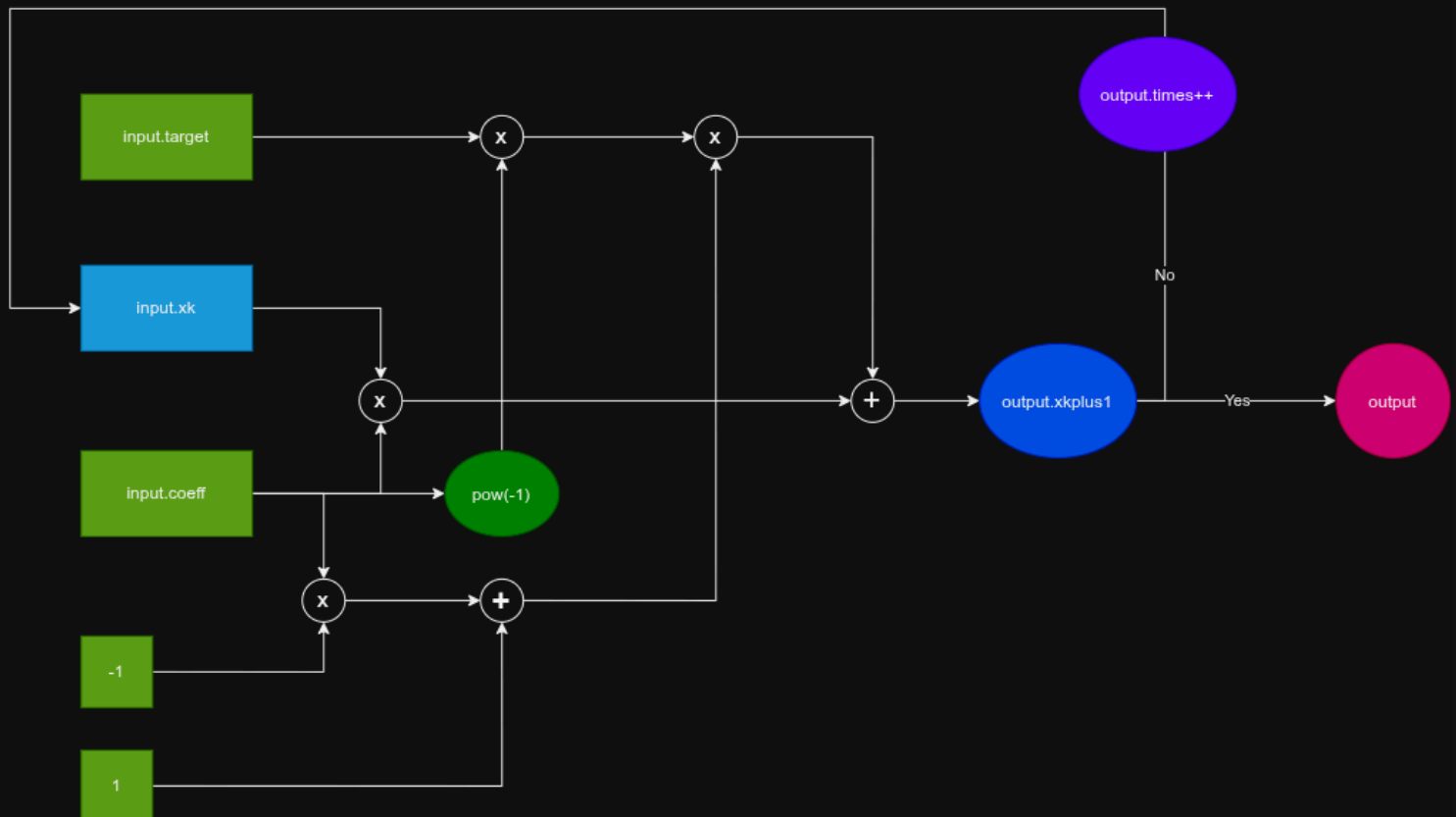


Recursive for Sqrt

Algorithm



Coding

```

#include<iostream>
#include<cmath>

using namespace std;

// constant
const int level = 12;
const double error = pow(10,-level);

// input and output structure
struct input{

```

C++


```
    output fin;

    // iteration
    fin.xkplus1 =
        (ini->coeff)*(ini->xk) +
        (1-(ini->coeff))*(ini->target)/(ini->xk);

    // test
    if(fabs((ini->xk)-(fin.xkplus1))<error){
        fin.times = 1;
        return fin;
    }
    else{
        ini->xk = fin.xkplus1;
        fin = Iter(ini);
        ++fin.times;
        return fin;
    }
}
```

Results

```
Enter a positive number to calculate the square root: 2
Enter an initial value that is greater than 0: 1.5
Enter a coefficient to iterate and enter 'q' to quit: 0.5
Result: 1.414213562373
Times: 5
Enter a coefficient to iterate and enter 'q' to quit: 0.6
Result: 1.414213562373
Times: 17
Enter a coefficient to iterate and enter 'q' to quit: 0.4
Result: 1.414213562373
Times: 17
Enter a coefficient to iterate and enter 'q' to quit: 0.3
Result: 1.414213562373
Times: 29
Enter a coefficient to iterate and enter 'q' to quit: 0.7
Result: 1.414213562374
Times: 28
Enter a coefficient to iterate and enter 'q' to quit: 0.8
Result: 1.414213562374
```

Console

